A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-30944890})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary ideal I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$I'^{\,(1-\sigma_x)^2}\,(t)\,J\mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 5851 & 2467 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,5851 + \mathbf{Z}\,(2467 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 30944890 = 0$, of discriminant -123779560. Class group invariants $(10 \quad 10 \quad 10 \quad 2)$, class number 2000; the three stored generator ideals (HNF) are $\begin{pmatrix} 199 & 159 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 193 & 99 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 157 & 81 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 6317)\, y^3 + (-6s - 354679)\, y^2 + (863s + 3255502)\, y + (-6552s + 9323580)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 12635*y^8 - 721992*y^7 + 78069741*y^6 - 4097539650*y^
5 + 114611694049*y^4 - 2106485720276*y^3 + 29464343022134*y^2 - 2892
42220002960*y + 1415353167138960
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{68572580114522542\,51} + \left(-\frac{418447}{6857258011452254251}\right) s, \qquad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5760833231840 = 2^5 \cdot 5 \cdot 23 \cdot 1565443813$:

$$\begin{pmatrix}
720104153980 & 635798831956 & 88527053297 & 519937678167 & 482354874216 & 505693212416 & 421856546838 & 631834141300 & 375408409468 & 620188178820 \\
0 & 4 & 0 & 0 & 2 & 0 & 2 & 0 & 0 & 2 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 314 factors with signed exponents of absolute value up to 277861696774991; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 1 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 6317)\, y^3 + (-6s - 354679)\, y^2 + (863s + 3255502)\, y + (-6552s + 9323580)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 12635*y^8 - 721992*y^7 + 78069741*y^6 - 4097539650*y^
5 + 114611694049*y^4 - 2106485720276*y^3 + 29464343022134*y^2 - 2892
42220002960*y + 1415353167138960
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849}\right) s, \qquad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $43807961796496 = 2^4 \cdot 23^2 \cdot 14891 \cdot 347579$:

$$\begin{pmatrix}
476173497788 & 202281566250 & 263495035516 & 385649339052 & 30691944248 & 367716659449 & 240380892912 & 18581880099 & 15112052413 & 11504705617 \\
0 & 2 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 46 & 13 & 24 & 15 & 4 & 31 & 27 & 29 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 267 factors with signed exponents of absolute value up to 234989190205813; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 6317)\, y^3 + (-6s - 354679)\, y^2 + (863s + 3255502)\, y + (-6552s + 9323580)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 12635*y^8 - 721992*y^7 + 78069741*y^6 - 4097539650*y^
5 + 114611694049*y^4 - 2106485720276*y^3 + 29464343022134*y^2 - 2892
42220002960*y + 1415353167138960
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + (-\frac{1403}{10368641602001000})\, s, \qquad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $55535605473200 = 2^4 \cdot 5^2 \cdot 83 \cdot 1672759201$:

$$\begin{pmatrix}
1388390136830 & 231626424108 & 290168754980 & 267053079090 & 26361350772 & 314677876133 & 1158620142392 & 575751006125 & 724748895157 & 237069318779 \\
0 & 2 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 10 & 4 & 3 & 8 & 0 & 6 & 2 & 2 \\
0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 262 factors with signed exponents of absolute value up to 273806086478056; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 350y^3 - 36sy^2 - 2232216y + 1296s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 700*y^8 - 4341932*y^6 + 38542026240*y^4 + 2095258694976*y^2 +
51975532362240
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 2 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + (-\frac{418447}{6857258011452254251})\, s, \qquad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10164350346390 = 2 \cdot 3^2 \cdot 5 \cdot 23 \cdot 30089 \cdot 163193$:

$$\begin{pmatrix}
3388116782130 & 738974532968 & 852479789800 & 602058177221 & 2028816217100 & 1254925282458 & 3218844804028 & 1908532760831 & 1990844299489 & 556095569316 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 202 factors with signed exponents of absolute value up to 194534772491075; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 350y^3 - 36sy^2 - 2232216y + 1296s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 700*y^8 - 4341932*y^6 + 38542026240*y^4 + 2095258694976*y^2 +
51975532362240
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 2 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + (\frac{113020}{2466815179409474849})\, s, \qquad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $373678321798385 = 5 \cdot 43 \cdot 47^2 \cdot 2089 \cdot 376639$:

$$\begin{pmatrix} 7950602591455 & 7676350054770 & 2386961699267 & 3729135836463 & 1948513309673 & 1209063530838 & 580293353483 & 3686019349724 & 6655536286429 & 5484826714423 \\ 0 & 47 & 42 & 42 & 1 & 16 & 0 & 2 & 16 & 10 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 220 factors with signed exponents of absolute value up to 73660983590830; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 350y^3 - 36sy^2 - 2232216y + 1296s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 700*y^8 - 4341932*y^6 + 38542026240*y^4 + 2095258694976*y^2 +
51975532362240
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 2 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + (-\frac{1403}{10368641602001000})\, s, \qquad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $69376616559000 = 2^3 \cdot 3^4 \cdot 5^3 \cdot 23 \cdot 41^2 \cdot 22153$:

$$\begin{pmatrix} 3133541850 & 2301356076 & 227538962 & 915124849 & 800847911 & 1340373220 & 2072635704 & 1693879079 & 2236123846 & 74397667 \\ 0 & 123 & 0 & 87 & 10 & 45 & 66 & 66 & 57 & 113 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 1 & 4 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 207 factors with signed exponents of absolute value up to 391557511845473; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s + 1184)\, y^3 + (29s - 134990)\, y^2 + (-575s + 735135)\, y + (8035s + 15013550)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2368*y^8 - 269980*y^7 + 33817016*y^6 - 2084432840*y^5 + 81574
375770*y^4 - 1692216124700*y^3 + 21139466582175*y^2 - 26386454766400
0*y + 2223246689492750
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + \left(-\frac{418447}{6857258011452254251}\right) s, \qquad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $19872590 = 2 \cdot 5 \cdot 1381 \cdot 1439$:

$$\begin{pmatrix} 19872590 & 7218944 & 6505022 & 5755294 & 4812106 & 14270977 & 18060912 & 3746316 & 16134451 & 8561617 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 270 factors with signed exponents of absolute value up to 9465277282274512; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s + 1184)\, y^3 + (29s - 134990)\, y^2 + (-575s + 735135)\, y + (8035s + 15013550)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2368*y^8 - 269980*y^7 + 33817016*y^6 - 2084432840*y^5 + 81574
375770*y^4 - 1692216124700*y^3 + 21139466582175*y^2 - 26386454766400
0*y + 2223246689492750
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849}\right) s, \qquad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1078714338834340 = 2^2 \cdot 5 \cdot 13^2 \cdot 23 \cdot 179 \cdot 77519129$:

$$\begin{pmatrix} 41489013032090 & 17509552991324 & 20516276877834 & 20374718940929 & 17464744551231 & 17054968657267 & 19338136669172 & 37522605803983 & 18402342521393 & 16514873231059 \\ 0 & 2 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 13 & 0 & 2 & 11 & 1 & 11 & 4 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 275 factors with signed exponents of absolute value up to 11667307327937480; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s + 1184)\, y^3 + (29s - 134990)\, y^2 + (-575s + 735135)\, y + (8035s + 15013550)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2368*y^8 - 269980*y^7 + 33817016*y^6 - 2084432840*y^5 + 81574
375770*y^4 - 1692216124700*y^3 + 21139466582175*y^2 - 26386454766400
0*y + 2223246689492750
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + (-\frac{1403}{10368641602001000})\, s, \qquad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $303301286209190 = 2 \cdot 5 \cdot 30330128620919$:

$$\begin{pmatrix} 303301286209190 & 144535334070024 & 293330223000494 & 257268800455997 & 44281878398431 & 259087006898581 & 203611577663471 & 161894452910022 & 257528713041066 & 294738176462577 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 264 factors with signed exponents of absolute value up to 8353534204799890; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2731y^3 - 16sy^2 - 1232700y + 2720s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5462*y^8 + 4992961*y^6 + 1188884440*y^4 - 1173893935600*y^2 +
228942674176000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{68572580114522542 51} + \left(-\frac{418447}{6857258011452254251}\right) s, \qquad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $204896335075288 = 2^3 \cdot 17^2 \cdot 257 \cdot 1229 \cdot 280583$:

$$\begin{pmatrix} 3013181398166 & 2743551611550 & 1602475580882 & 2147121121112 & 2583422040399 & 2888272425438 & 2491093196943 & 30956552698 & 2623692094934 & 1381786880632 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 17 & 1 & 0 & 4 & 10 & 4 & 16 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 280 factors with signed exponents of absolute value up to 14775808026; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2731y^3 - 16sy^2 - 1232700y + 2720s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5462*y^8 + 4992961*y^6 + 1188884440*y^4 - 1173893935600*y^2 +
228942674176000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849}\right) s, \qquad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $121258264420080 = 2^4 \cdot 3^2 \cdot 5 \cdot 168414256139$:

$$\begin{pmatrix}
10104855368340 & 9345938218092 & 1736754698582 & 155971325962 & 8746527377753 & 8245274824972 & 7634551891357 & 4228615021960 & 1329977864360 & 7969642513744 \\
0 & 6 & 2 & 1 & 3 & 2 & 4 & 1 & 4 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 295 factors with signed exponents of absolute value up to 9082335405; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2731y^3 - 16sy^2 - 1232700y + 2720s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5462*y^8 + 4992961*y^6 + 1188884440*y^4 - 1173893935600*y^2 +
228942674176000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + (-\frac{1403}{10368641602001000})\, s, \qquad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $76340748047450 = 2 \cdot 5^2 \cdot 157 \cdot 9724936057$:

$$\begin{pmatrix} 15268149609490 & 4015539492132 & 15143472830740 & 4176181107517 & 8580765040692 & 4650534239978 & 10347153147842 & 12899515824097 & 5540278246869 & 7670252068737 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 4 & 0 & 3 & 4 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 292 factors with signed exponents of absolute value up to 9371789712; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 796)\, y^3 + (29s - 86202)\, y^2 + (-259s + 3391035)\, y + (-2145s - 26272620)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 1596*y^8 - 175588*y^7 + 38705384*y^6 - 1998146584*y^5
+ 54988508514*y^4 - 958554568660*y^3 + 14254583550895*y^2 - 1437995
71195500*y + 832628774176650
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + (-\frac{418447}{6857258011452254251})\, s, \qquad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $51391995874488 = 2^3 \cdot 3^2 \cdot 569 \cdot 1254442391$:

$$\begin{pmatrix} 4282666322874 & 2151242738856 & 2953408694610 & 2985711139966 & 3605334430085 & 3463802993073 & 2217459160528 & 1187121733047 & 1684702554658 & 1574334602152 \\ 0 & 6 & 2 & 1 & 4 & 4 & 1 & 2 & 4 & 2 \\ 0 & 0 & 2 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 287 factors with signed exponents of absolute value up to 3901317161540652; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 3 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 796)\, y^3 + (29s - 86202)\, y^2 + (-259s + 3391035)\, y + (-2145s - 26272620)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 1596*y^8 - 175588*y^7 + 38705384*y^6 - 1998146584*y^5
+ 54988508514*y^4 - 958554568660*y^3 + 14254583550895*y^2 - 1437995
71195500*y + 832628774176650
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849}\right) s, \qquad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $43176275936000 = 2^8 \cdot 5^3 \cdot 149 \cdot 2029 \cdot 4463$:

$$\begin{pmatrix}
53970344920 & 45391638180 & 42672553330 & 11800538470 & 20085537740 & 11917797526 & 25664641930 & 22929394091 & 7784409348 & 30292019252 \\
0 & 4 & 0 & 2 & 0 & 1 & 1 & 0 & 1 & 3 \\
0 & 0 & 10 & 0 & 0 & 9 & 7 & 2 & 9 & 1 \\
0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 10 & 3 & 2 & 2 & 0 & 2 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 286 factors with signed exponents of absolute value up to 2861328727268913; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 796)\, y^3 + (29s - 86202)\, y^2 + (-259s + 3391035)\, y + (-2145s - 26272620)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 1596*y^8 - 175588*y^7 + 38705384*y^6 - 1998146584*y^5
+ 54988508514*y^4 - 958554568660*y^3 + 14254583550895*y^2 - 1437995
71195500*y + 832628774176650
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + (-\frac{1403}{10368641602001000})\,s, \qquad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $35172728358046 = 2 \cdot 1979 \cdot 8886490237$:

$$\begin{pmatrix} 35172728358046 & 2763587212181 & 24506509926714 & 31604259146119 & 32995168233535 & 14910789804599 & 10236947070795 & 10276446727203 & 15105446998558 & 19110838546860 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 290 factors with signed exponents of absolute value up to 1202361373399609; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1361y^3 - 2sy^2 + 205690y + 200s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2722*y^8 + 2263701*y^6 - 436108620*y^4 + 17552464100*y^2 + 12
37795600000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1199530329}{6857258011452254251} + (-\frac{418447}{6857258011452254251})\,s, \qquad J = \begin{pmatrix} 5851 & 2467 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5851$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $117296673946304 = 2^6 \cdot 1832760530411$:

$$\begin{pmatrix} 3665521060822 & 1031049478788 & 2881010444528 & 2855985938460 & 1287594759918 & 1936714745841 & 1291932329037 & 1705565258488 & 838021523557 & 3195614264611 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 300 factors with signed exponents of absolute value up to 515249827265; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1361y^3 - 2sy^2 + 205690y + 200s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2722*y^8 + 2263701*y^6 - 436108620*y^4 + 17552464100*y^2 + 12
37795600000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{1439284543}{2466815179409474849} + \left(\frac{113020}{2466815179409474849}\right) s, \qquad J = \begin{pmatrix} 4769 & 3484 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4769$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $292322561024000 = 2^{13} \cdot 5^3 \cdot 13^2 \cdot 883 \cdot 1913$:

$$\begin{pmatrix} 1756746160 & 1233405680 & 1697379940 & 207087290 & 257914106 & 680054070 & 1279015750 & 190737176 & 803020491 & 195358384 \\ 0 & 520 & 160 & 250 & 116 & 8 & 184 & 481 & 63 & 429 \\ 0 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 10 & 6 & 8 & 2 & 7 & 6 & 7 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 295 factors with signed exponents of absolute value up to 157065219523; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1361y^3 - 2sy^2 + 205690y + 200s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2722*y^8 + 2263701*y^6 - 436108620*y^4 + 17552464100*y^2 + 12
37795600000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{32701}{51843208010005} + \left(-\frac{1403}{10368641602001000}\right) s, \qquad J = \begin{pmatrix} 4010 & 220 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4010$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $53104303260000 = 2^5 \cdot 3^2 \cdot 5^4 \cdot 1049 \cdot 281243$:

$$\begin{pmatrix} 17701434420 & 2475966390 & 2897088930 & 12605418090 & 9610582418 & 3155480775 & 17238565915 & 9913437973 & 17652246375 & 11093366319 \\ 0 & 5 & 0 & 0 & 3 & 4 & 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 30 & 28 & 24 & 18 & 14 & 0 & 13 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 309 factors with signed exponents of absolute value up to 403594772202; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.